

SESSION 3 — Bar Charts

What is a Bar Chart?

A bar chart is used to represent **categorical data using rectangular bars**.

Why Use Bar Charts?

Bar charts are useful when:

- Comparing values across categories
 - Showing differences clearly
 - Ranking categories
-

Examples

- Revenue by Region
 - Sales by Product
 - Students per department
-

Types of Bar Charts

1. Vertical Bar Chart → `plt.bar()`
2. Horizontal Bar Chart → `plt.barh()`

✓ `plt.bar()`

Syntax

```
plt.bar(x, height, width=0.8, color=None, label=None)
```

Parameters

`x` → categories (x-axis)
`height` → values (y-axis)

width (default = 0.8)

- Controls bar thickness

color (default = None)

- Bar color

label (default = None)

- Used for legend

Use Case

- Compare values across categories

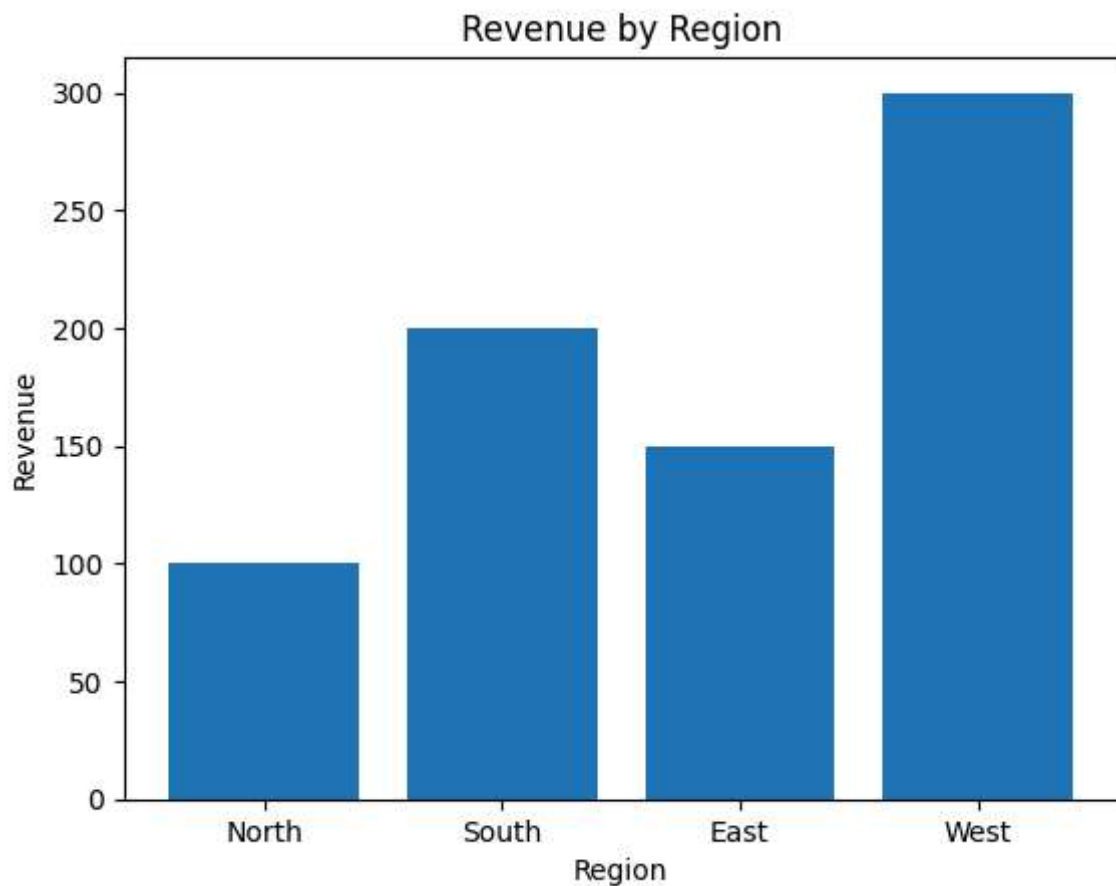
```
import matplotlib.pyplot as plt

regions = ["North","South","East","West"]
revenue = [100,200,150,300]

plt.bar(regions, revenue)

plt.title("Revenue by Region")
plt.xlabel("Region")
plt.ylabel("Revenue")

plt.show()
```



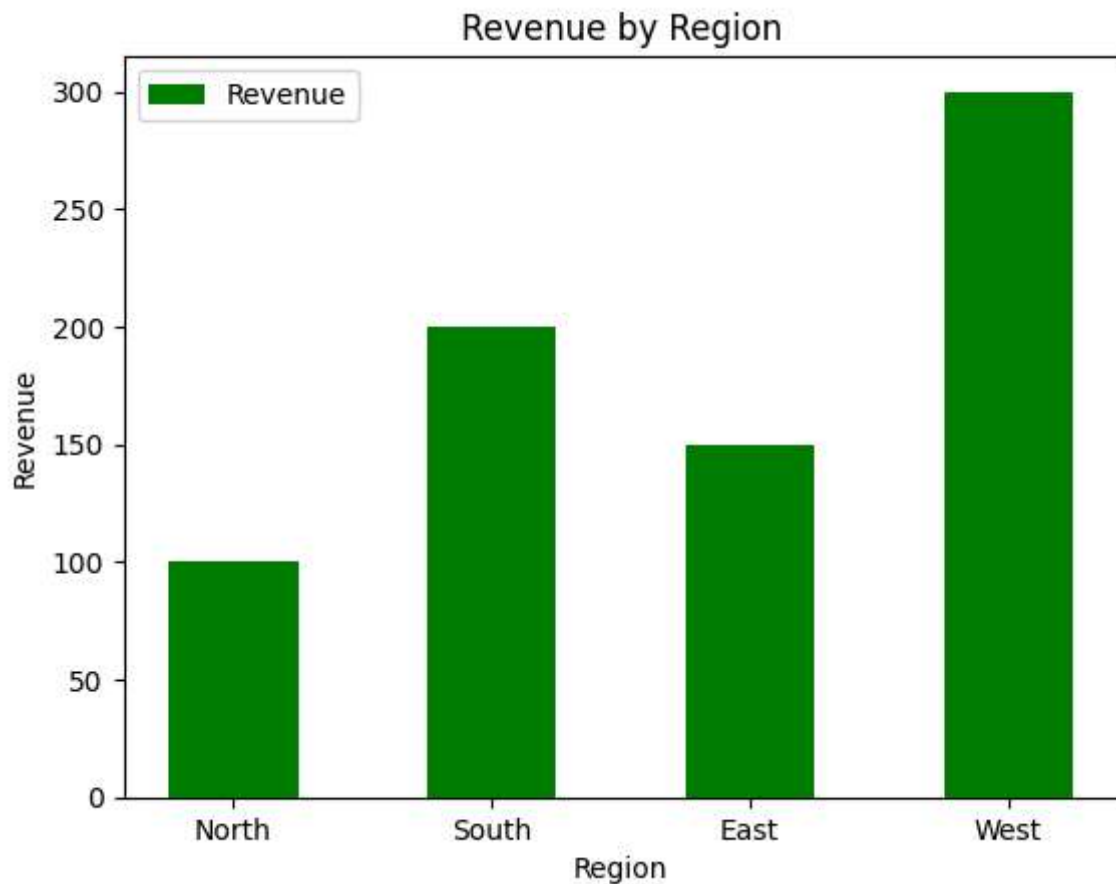
✓ Customizing Bar Chart

We can modify:

- Color
- Width
- Labels

```
plt.bar(  
    regions,  
    revenue,  
    color='green',  
    width=0.5,  
    label="Revenue"  
)  
  
plt.title("Revenue by Region")  
plt.xlabel("Region")  
plt.ylabel("Revenue")  
  
plt.legend()
```

```
plt.show()
```



✓ Adding Text Labels on Bars

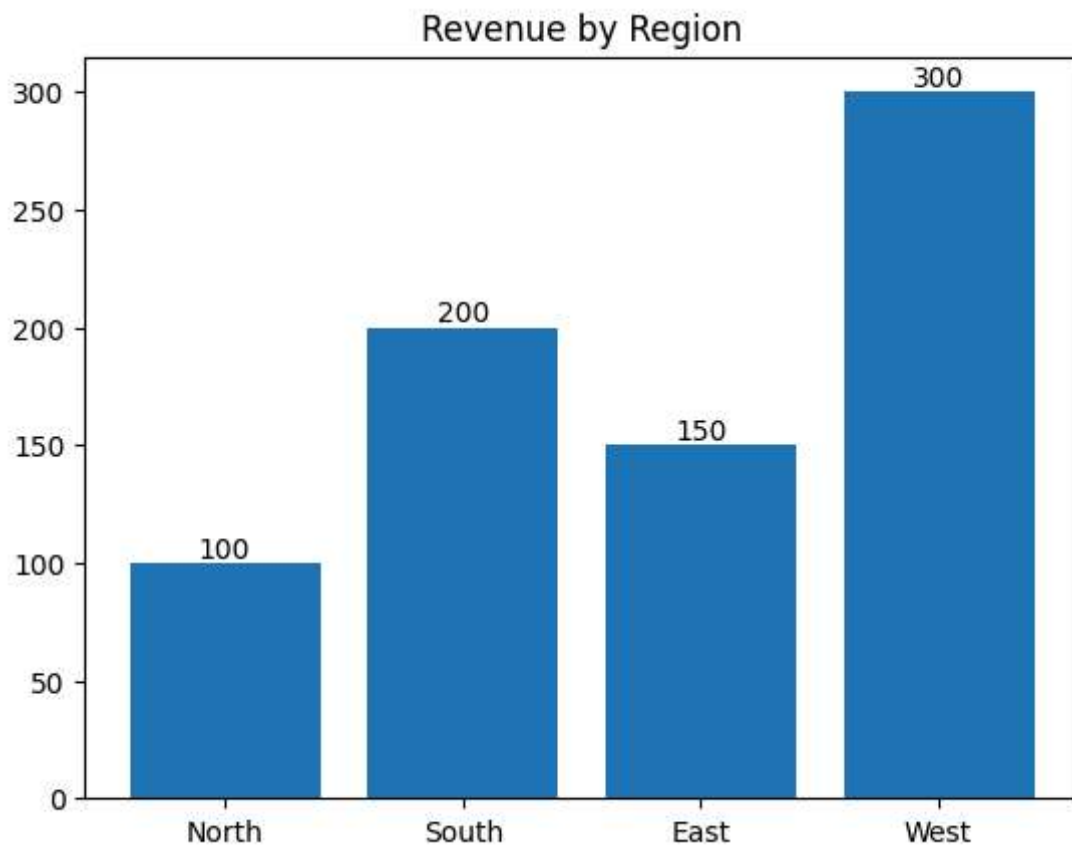
Why?

To display exact values on each bar.

```
bars = plt.bar(regions, revenue)

plt.bar_label(bars)

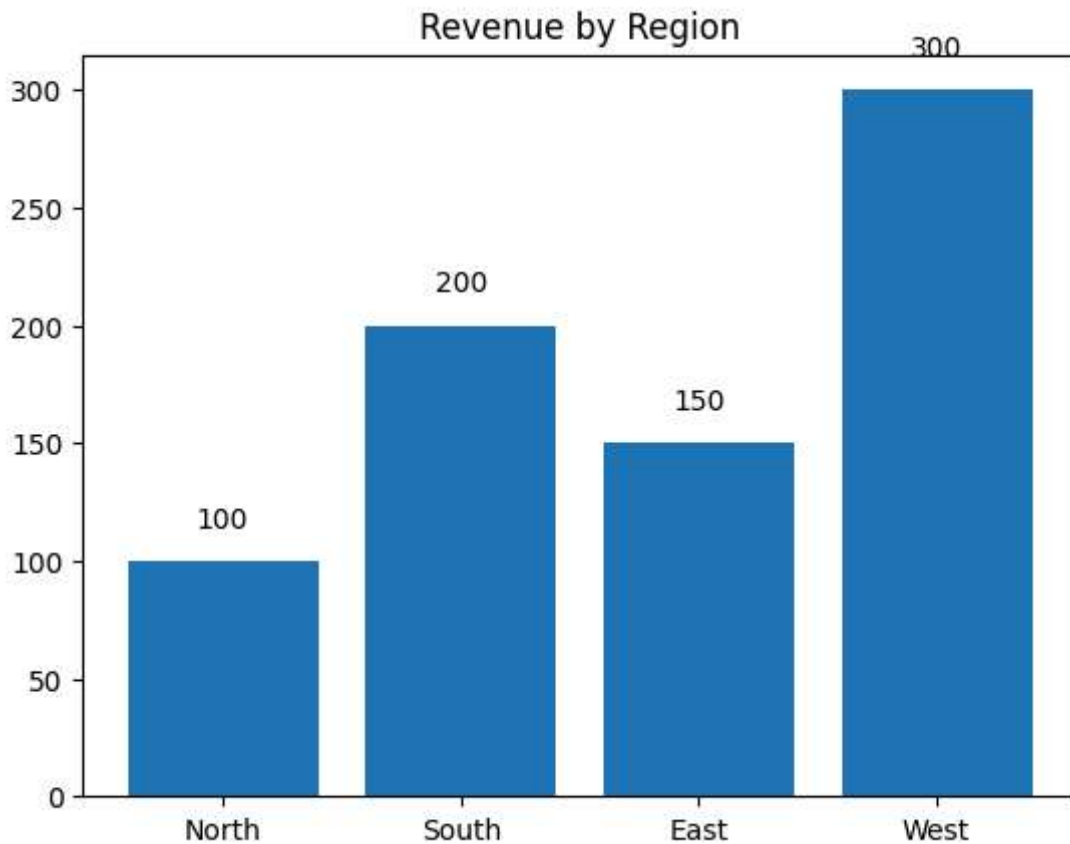
plt.title("Revenue by Region")
plt.show()
```



```
bars = plt.bar(regions, revenue)

plt.bar_label(bars, padding=10)

plt.title("Revenue by Region")
plt.show()
```



✓ Horizontal Bar Chart — `plt.barh()`

Purpose

Used when:

- Category names are long
- Better readability is required

Syntax

```
plt.barh(y, width, height=0.8)
```

Parameters

`y` → categories

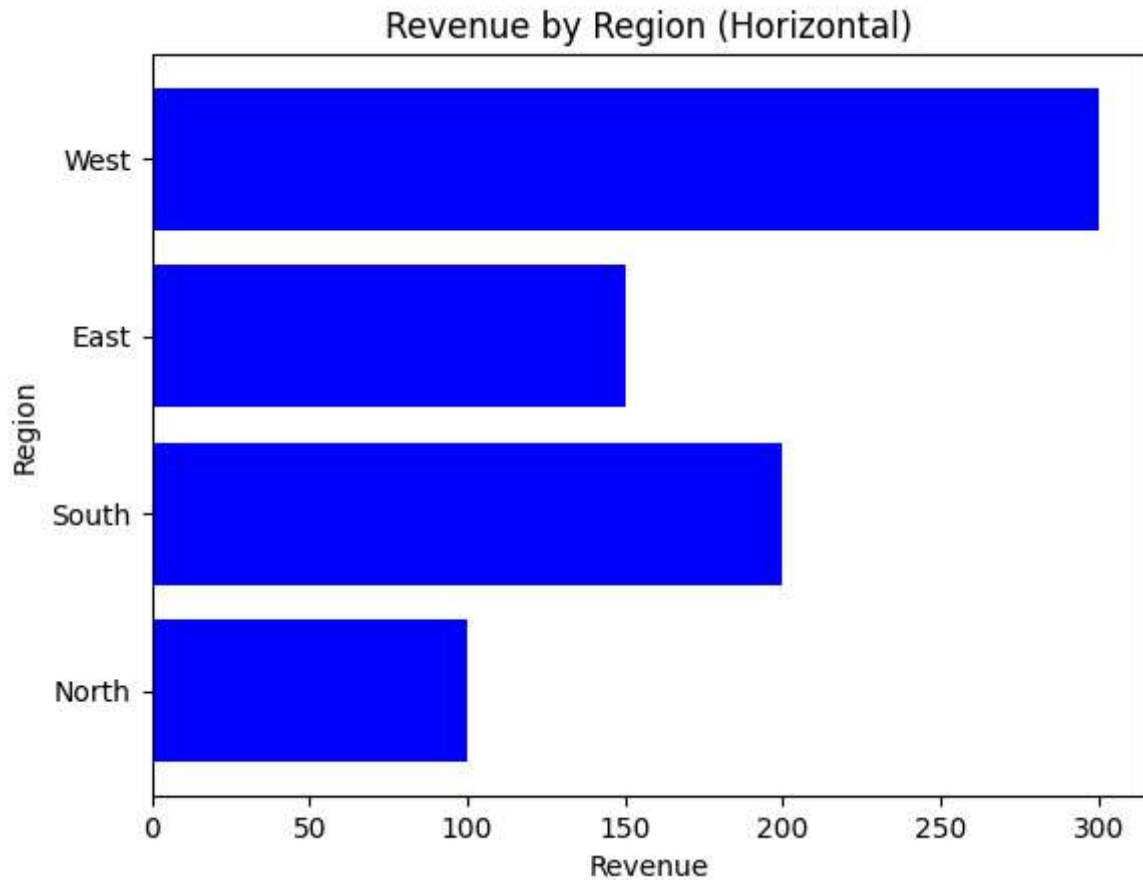
`width` → values

`height` (default = 0.8) → bar thickness

```
plt.barh(regions, revenue, color='blue')

plt.title("Revenue by Region (Horizontal)")
plt.xlabel("Revenue")
plt.ylabel("Region")

plt.show()
```

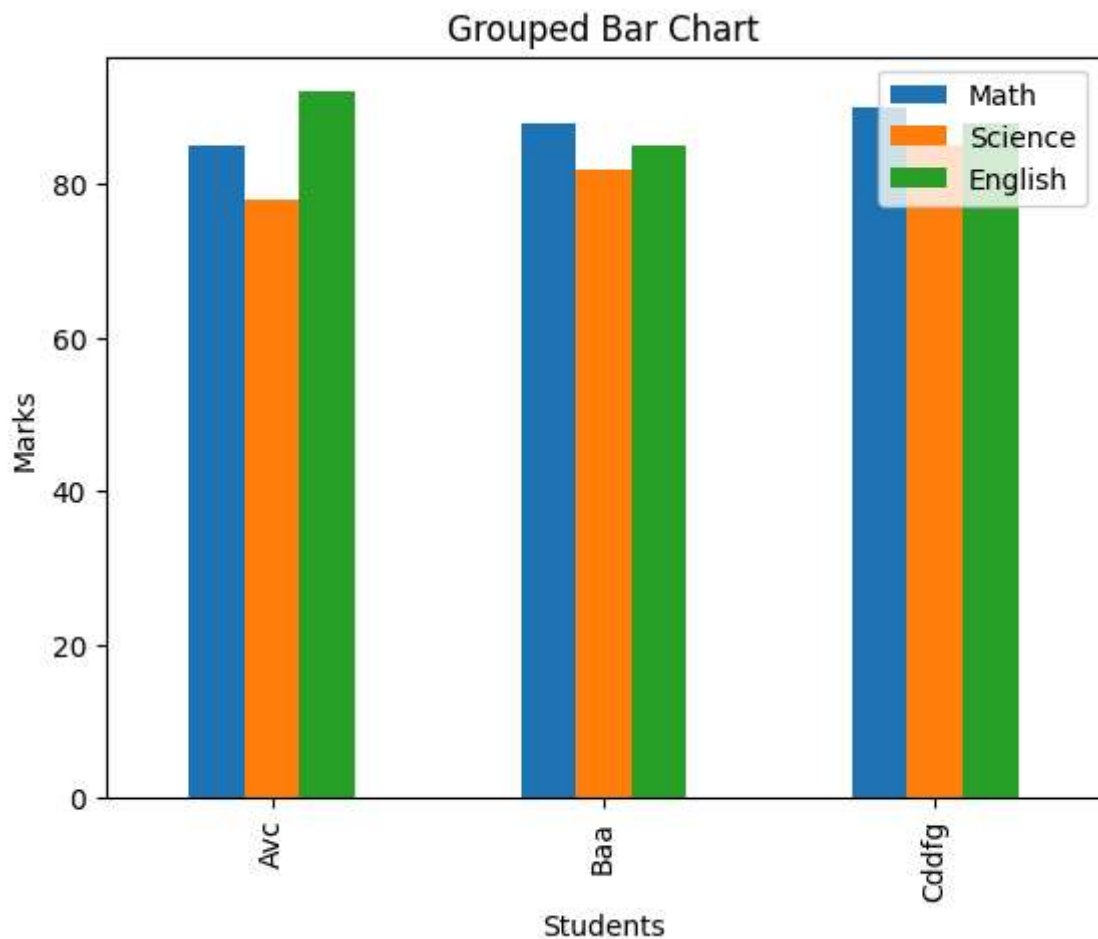


```
import pandas as pd
import matplotlib.pyplot as plt

# Create DataFrame
df = pd.DataFrame({
    'Math': [85, 88, 90],
    'Science': [78, 82, 85],
    'English': [92, 85, 88]
}, index=['Avc', 'Baa', 'Cddfg'])

# Plot grouped bar chart
df.plot(kind='bar')

plt.xlabel('Students')
plt.ylabel('Marks')
plt.title('Grouped Bar Chart')
plt.show()
```



```
import numpy as np
import matplotlib.pyplot as plt

# Data
students = ['A', 'B', 'C']
math_marks = [85, 88, 90]
science_marks = [78, 82, 85]
english_marks = [92, 85, 88]

# X positions
x = np.arange(len(students)) # [0,1,2]
width = 0.1 # width of bars

# Create bars
plt.bar(x - width, math_marks, width, label='Math')
plt.bar(x, science_marks, width, label='Science')
plt.bar(x + width, english_marks, width, label='English')

# Labels and Title
plt.xlabel('Students')
plt.ylabel('Marks')
plt.title('Grouped Bar Chart - Student Marks')

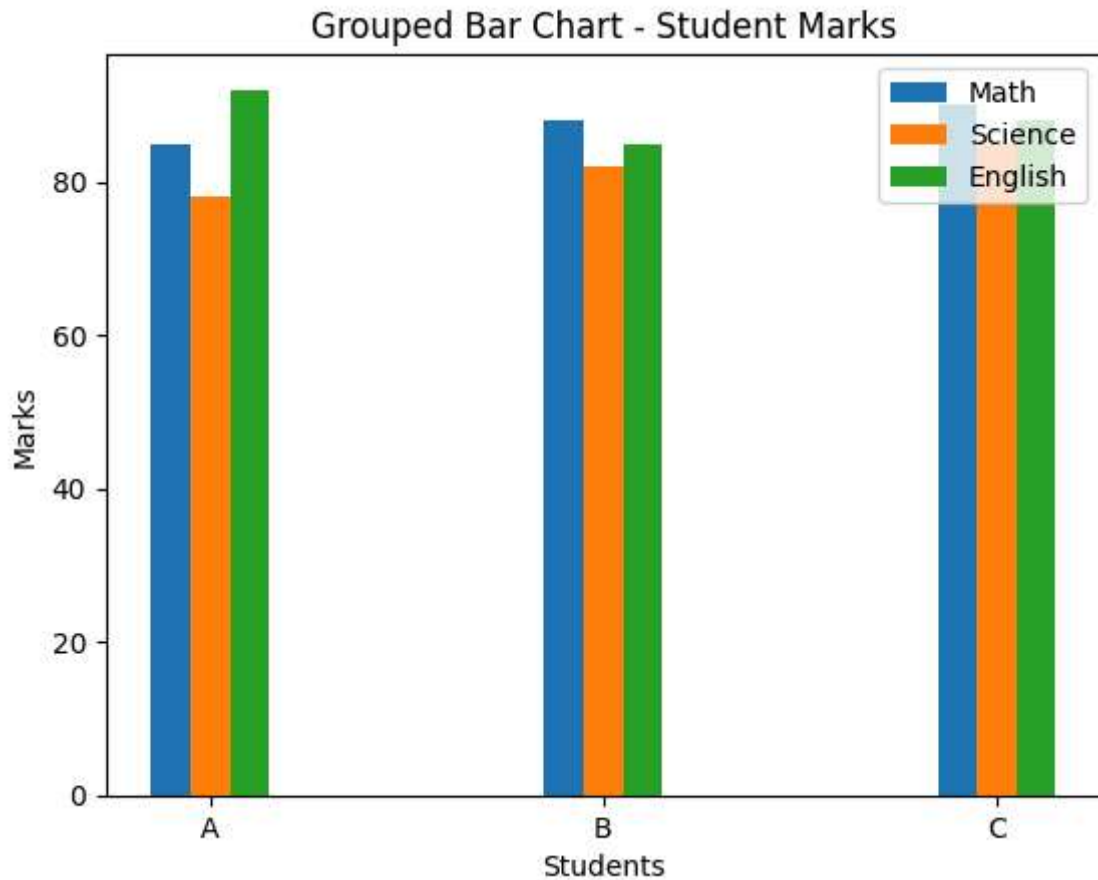
# Ticks
```



```
plt.xticks(x, students)

# Legend
plt.legend()

# Show plot
plt.show()
```



Pie Chart

What is a Pie Chart?

A pie chart shows **proportions of categories** as parts of a whole.

Why Use Pie Chart?

- To show percentage contribution
- To visualize composition

Example

- Market share
- Budget allocation

✓ plt.pie()

Syntax

```
plt.pie(x, labels=None, autopct=None, explode=None)
```

Parameters

$x \rightarrow$ values

labels (default=None)

- Category names

autopct (default=None)

- Displays percentage

Example: '%1.1f%%'

explode (default=None)

- Highlights slice

Example: [0.1, 0, 0, 0]

Use Case

- Show percentage distribution

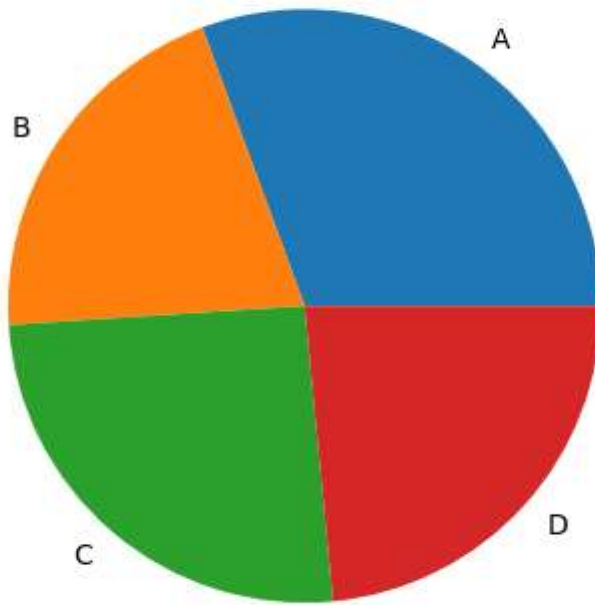
```
labels = ["A", "B", "C", "D"]
values = [30, 20, 25, 23]

plt.pie(values, labels=labels)

plt.title("Basic Pie Chart")

plt.show()
```

Basic Pie Chart



Pie Chart with Percentage and Explode

✓ Explanation of autopct and explode in plt.pie()

1. autopct

Purpose

Used to display **percentage values on each slice of the pie chart**.

Syntax Example

```
autopct='%1.1f%%'
```

Meaning of '%1.1f%%'

Breakdown:

- % → formatting starts
- 1.1f → float number with 1 decimal place
- %% → literal % symbol

Example

If a slice represents 25%:

It will display as:

25.0%

Variations

- '%1.0f%%' → no decimal (25%)
 - '%1.2f%%' → two decimals (25.00%)
-

2. explode

Purpose

Used to **separate (highlight) specific slices** of the pie chart.

Syntax Example

```
explode = [0.1, 0, 0, 0]
```

How It Works

Each value in the list corresponds to each slice:

- 0 → no separation
 - >0 → slice moves outward
-

Example

```
explode = [0.1, 0, 0, 0]
```

- First slice moves slightly outward
 - Others remain in place
-

Use Case

- Highlight important category
 - Emphasize highest value
-

Summary

Parameter	Purpose
autopct	Show percentage on chart
explode	Highlight specific slice

